
Qwen-Audio-3.1-ASR Technical Report

Qwen-Audio ASR Team

 **Alibaba Token Foundry**

 [Project Page](#)

Abstract

We present **Qwen-Audio-3.1-ASR**, a family of audio-language models that builds upon Qwen-Audio-3.0-ASR with improved speech recognition and expanded capabilities in speaker-diarized transcription and general audio understanding. The family retains support for 30 languages and 16 Chinese dialectal varieties, together with instruction-controlled capabilities including historical-context conditioning, hierarchical hotword customization, and native transcription polishing. Beyond conventional transcription, speaker-diarized ASR produces structured outputs containing speaker labels, timestamps, and recognized text, supporting speaker attribution across turns and the representation of overlapping speech. The general audio understanding capabilities extend to audio, speech, and music captioning, temporal grounding and sound event detection, as well as audio question answering and multiple-choice reasoning. These tasks broaden the scope of the system from recognizing spoken words to interpreting speakers, acoustic events, and their temporal relationships. To accommodate different deployment requirements, the family includes variants for asynchronous file transcription, low-latency streaming, interactive voice input, and non-streaming recognition, with speaker-diarized transcription and general audio understanding available in the corresponding variants. Evaluations cover conventional ASR benchmarks, multilingual and dialectal speech, speaker-diarized transcription, and general audio understanding, demonstrating the capabilities of the Qwen-Audio-3.1-ASR family across transcription and broader audio analysis tasks.

1 Introduction

Automatic speech recognition (ASR) provides the linguistic foundation for voice interfaces, meeting transcription, and conversational systems. However, a transcript alone captures only part of the information in an audio recording. Multi-party conversations require identifying who spoke and when, while interpreting speech, environmental sounds, and music requires information that may not be recoverable from words alone. Practical audio systems must therefore address three complementary questions: *what was said*, *who said it and when*, and *what the audio conveys beyond its transcript*. These requirements motivate extending accurate speech recognition toward structured transcription and general audio understanding.

Recent audio-language models have made substantial progress toward this broader scope. Qwen3-ASR demonstrates the potential of building multilingual speech recognition upon an audio-capable language model [1]. In parallel, Qwen3-Omni integrates audio perception into a broader multimodal framework [2], while MOSS-Audio explicitly supports captioning, timestamped transcription, and time-aware audio understanding [3]. More recently, FireRedAudio combines recognition and audio understanding with speech generation in a general-purpose audio-language model [4]. Together, these

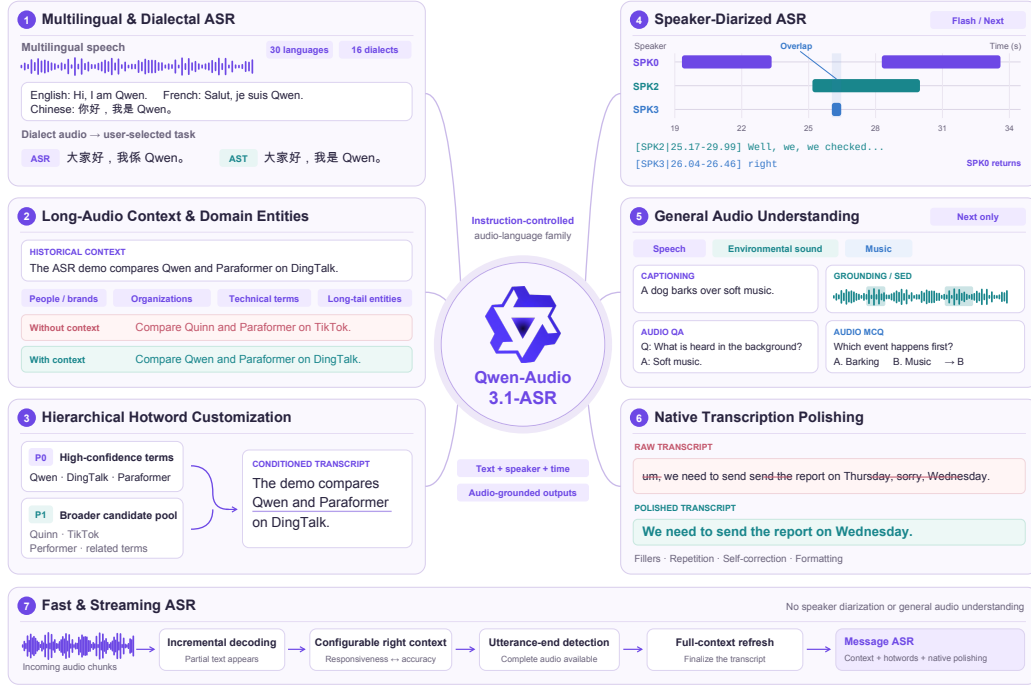


Figure 1: **Overview of Qwen-Audio-3.1-ASR and its capabilities.** The model family supports multilingual and dialectal recognition, user-selectable dialect transcription or translation into Mandarin, long-audio context and domain entity recognition, hierarchical hotword customization, and native transcription polishing. Speaker-diarized ASR produces transcripts with speaker labels and timestamps, including overlapping speech, while general audio understanding supports captioning, temporal grounding, sound event detection, and audio question answering. Streaming recognition provides incremental transcripts for real-time display, followed by a non-streaming refresh at utterance end. Capability availability depends on the model variant; streaming does not support speaker diarization or general audio understanding.

developments establish audio-conditioned language modeling as a useful framework for tasks that range from lexical recognition to descriptions and answers grounded in acoustic input.

Speaker-diarized transcription introduces a distinct structural requirement. Beyond recognizing individual utterances, a system must associate words with consistent speaker labels and temporal boundaries, including during rapid turn-taking and overlapping speech. Recent systems such as MOSS Transcribe Diarize and VibeVoice-ASR formulate transcription, speaker attribution, and timestamp prediction jointly, demonstrating an alternative to assembling these outputs through separate processing stages [5, 6]. This direction is particularly relevant to meetings and interviews, where a lexically accurate transcript can still misrepresent a conversation if speaker assignments or temporal relationships are incorrect.

General audio understanding further broadens the information that a model must preserve. Environmental events, vocal expressions, and musical characteristics can be central to an answer even when they have no direct textual transcription. Moreover, identifying an event is not equivalent to locating it in time or reasoning about its relationship to other sounds. Benchmarks such as MMAU evaluate understanding and reasoning across speech, sound, and music [7], while MMSU examines fine-grained spoken-language understanding that involves semantic, paralinguistic, and phonological information [8]. These complementary perspectives motivate evaluating audio models beyond transcription error rates alone.

The challenge for a production-oriented model family is to develop these capabilities while retaining reliable recognition and practical control over its outputs. Multilingual and dialectal speech, specialized terminology, and evolving conversational context continue to place demands on ASR accuracy.

Speaker-aware outputs additionally require consistent attribution and timing, whereas audio understanding requires responses grounded in the recording rather than plausible text alone. Deployment also imposes different constraints: low-latency streaming, asynchronous file transcription, and richer non-streaming analysis need not expose identical task capabilities. We therefore treat recognition quality, output structure, audio understanding, and deployment configuration as related but distinct aspects of the system.

In this report, we present **Qwen-Audio-3.1-ASR**, a family of audio-language models that builds upon Qwen-Audio-3.0-ASR with stronger speech recognition and two major capability extensions: speaker-diarized ASR and general audio understanding. The family retains support for 30 languages and 16 Chinese dialectal varieties, together with production-oriented controls such as contextual conditioning, hierarchical hotwords, and native transcription polishing in the corresponding variants. Its non-streaming Flash and Next variants support speaker-diarized transcription, while Next additionally supports general audio understanding. This organization extends the scope of the family without implying that every deployment variant provides the same functionality.

The main advances of this generation are summarized as follows:

- **Enhanced speech recognition.** We strengthen the core ASR capability across multilingual, dialectal, and application-oriented recognition settings, building on the instruction-controlled recognition framework of the previous generation. Contextual conditioning, hotword customization, and transcription polishing remain complementary controls rather than substitutes for accurate recognition.
- **Speaker-diarized ASR.** We extend transcription to structured outputs containing recognized text, speaker labels, and timestamps. The capability targets multi-speaker recordings with speaker changes, short conversational responses, and overlapping speech, preserving conversational structure beyond a flat sequence of words.
- **General audio understanding.** We extend the Next variant beyond transcription to captioning of speech, environmental sounds, and music; temporal audio grounding; sound event detection; and audio question answering in open-ended and multiple-choice formats. These tasks connect acoustic evidence with semantic descriptions, temporal localization, and audio-grounded reasoning.

Our evaluation spans speech recognition, speaker-diarized transcription, and audio understanding, using task-appropriate measures to characterize these different capabilities. The remainder of this report describes the model architecture in Section 2, training in Section 3, deployment variants in Section 4, and experimental evaluations in Section 5.

2 Model Architecture

Qwen-Audio-3.1-ASR builds upon the audio-language modeling framework of Qwen-Audio-3.0-ASR and extends its output space from speech transcription to speaker-diarized transcription and general audio understanding. The overall framework consists of an audio encoder, an audio adapter, an LLM-based decoder, and an auxiliary CTC branch for recognition-related processing. Audio representations provide the acoustic evidence, while task instructions and optional textual context specify the desired output behavior. The capabilities available through this framework depend on the deployment variant.

2.1 Audio Encoding and Instruction-Conditioned Decoding

The audio encoder transforms the input waveform into a sequence of acoustic representations. The audio adapter bridges these representations to the input space of the language-model decoder, allowing the decoder to generate outputs conditioned on both audio and textual instructions. Unlike a transcript-only interface, this formulation retains direct access to the acoustic input when producing speaker-aware transcripts, audio descriptions, or answers to audio-related questions.

Let x denote the input audio, q the task instruction, and c the optional textual conditions. The conditional generation process can be expressed as

$$H = A(E(x)), \quad p_{\theta}(y \mid x, q, c) = \prod_{t=1}^T p_{\theta}(y_t \mid y_{<t}, H, q, c), \quad (1)$$

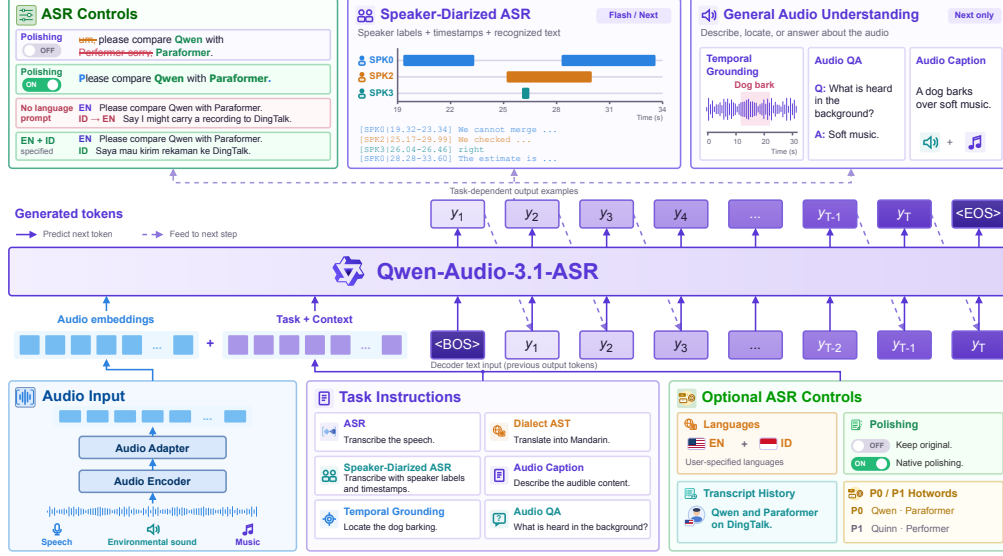


Figure 2: Overview of the instruction-controlled decoding workflow of Qwen-Audio-3.1-ASR.

where E is the audio encoder, A is the audio adapter, and y is the task-dependent output sequence. This formulation describes the conditioning relationships without prescribing a particular prompt template or output serialization.

For ASR, the textual conditions can include target languages, historical transcripts, hierarchical hotword candidates, and a polishing instruction. These controls specify how the recognized content should be rendered or contextualized. For audio understanding, the instruction instead specifies a description, a target event, or a question to be answered from the recording.

The auxiliary CTC branch operates on the audio encoder representations and provides an initial recognition hypothesis for hotword customization and a coarse alignment signal for long-audio context injection, as described in Sections 5.4 and 5.5. Its role is auxiliary: CTC hypotheses are not the sole representation of the recording, and should not be confused with the speaker-attributed timestamps or event boundaries produced by the corresponding tasks.

2.2 Speaker-Diarized Transcription

Speaker-diarized ASR extends the transcription target to include speaker attribution and temporal information. Instead of returning a single undifferentiated text sequence, the model produces a structured transcript that associates each speech segment with its recognized content, speaker label, and temporal boundaries.

At the task level, a speaker-diarized transcript can be represented as

$$\mathcal{S} = \{(b_i, e_i, k_i, w_i)\}_{i=1}^N, \quad (2)$$

where b_i and e_i are the segment start and end times, k_i is a speaker label, and w_i is the corresponding transcription. This notation specifies the semantic content of the output rather than an exact implementation format. Speaker labels identify distinct speakers within a recording; they do not imply recognition of a person’s real-world identity.

The output must preserve speaker consistency across turns while accommodating short responses and overlapping speech. In particular, temporal intervals associated with different speakers need not be disjoint. Jointly representing text, speaker attribution, and timing allows the transcript to retain conversational structure that would otherwise be lost in plain-text ASR output.

2.3 General Audio Understanding

The Next variant broadens instruction-conditioned generation beyond lexical transcription to comprehensive audio understanding across speech, environmental sounds, and music. Specifically, its training incorporates three complementary task families within a unified framework: audio captioning, temporal grounding, and audio question answering.

Audio captioning. Audio captioning provides natural-language descriptions of audio content and its associated attributes. Depending on the audio type, these attributes may include speaker and paralinguistic characteristics, sound sources and acoustic environments, or musical properties such as genre, instrumentation, rhythm, and mood.

Temporal grounding. Temporal grounding associates a semantic query, such as a fine-grained caption or a sound event category, with its corresponding temporal locations in audio. We consider two grounding tasks: temporal boundary localization predicts the start and end boundaries of a single target interval, whereas temporal retrieval identifies all time intervals that match the query. This formulation provides a unified view of single-interval localization and multi-interval retrieval.

Audio question answering (QA). Audio question answering further requires the model to interpret acoustic evidence in response to a given query. Both open-ended and multiple-choice formats are included, supporting the integration of audio perception with semantic reasoning.

2.4 Deployment Variants

The family exposes different capability combinations to meet different deployment requirements. The non-streaming Flash and Next variants support speaker-diarized ASR, while Next additionally supports general audio understanding. File-transcription, streaming, and message-oriented variants provide recognition-focused interfaces with their respective supported controls. Consequently, the task formulations above describe the scope of the family rather than a requirement that every variant support every task. The deployment configurations are detailed in Section 4.

3 Training

The training of Qwen-Audio-3.0-ASR comprises five stages: audio encoder pretraining, audio-language model pretraining, LLM cooldown adaptation, supervised fine-tuning, and reinforcement learning. These stages progressively align acoustic representations with linguistic knowledge and instruction-following capabilities, while further optimizing the model for practical ASR objectives.

3.1 Pretraining of Audio Encoder

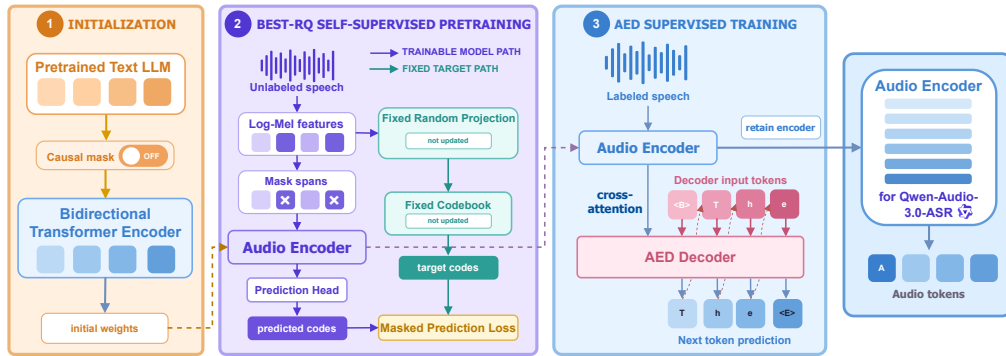


Figure 3: The pretraining pipeline for the audio encoder.

As illustrated in Figure 3, the audio encoder is trained in two consecutive stages: BEST-RQ self-supervised pretraining on large-scale unlabeled speech, followed by supervised training with an

attention-based encoder–decoder (AED) objective on labeled audio–text data. Across the two stages, the pretraining corpus comprises tens of millions of hours of audio collected from diverse real-world scenarios.

BEST-RQ self-supervised pretraining. We first train the encoder using BEST-RQ [9] on large-scale unlabeled speech collected from diverse real-world scenarios, covering domains such as artificial intelligence, biotechnology, e-commerce, education, entertainment, finance, and mobility. Acoustic frames are randomly masked, and the encoder predicts discrete targets generated from the corresponding unmasked features by a fixed random projection and a frozen codebook. This objective enables the encoder to learn contextual speech representations from large-scale and diverse unlabeled audio. Following our previous study on cross-modal initialization [10], we initialize compatible Transformer layers of the encoder from a pretrained Qwen3 LLM [11]. Causal attention is replaced with bidirectional attention for speech encoding, while the acoustic front end is initialized separately. The resulting network is subsequently adapted to speech through BEST-RQ pretraining.

Supervised AED training. The pretrained encoder is then integrated into an AED model and optimized on large-scale labeled audio–text data, primarily covering Chinese and English. The labeled data are processed through a comprehensive pipeline that includes voice activity detection (VAD), pseudo-label generation by multiple ASR systems, including Paraformer-V2 [12], Whisper, and SenseVoice [13], followed by inverse text normalization (ITN). Following the general training paradigm of SenseVoice-Large [13], an autoregressive decoder attends to the encoder representations and predicts the reference transcription token by token, providing direct acoustic–text supervision. After this stage, the AED decoder is discarded, and the trained encoder is used to initialize the audio encoder of Qwen-Audio-3.0-ASR for subsequent Audio LLM pretraining.

3.2 Pretraining of Audio LLM

Initialization. The Audio LLM decoder is initialized from the language-model component of Qwen, while its vision-specific components are not used. The backbone adopts a Mixture-of-Experts (MoE) architecture. This provides substantial linguistic capacity and multilingual knowledge while maintaining relatively low per-token computation, making it suitable for Qwen-Audio-3.0-ASR, including its latency-sensitive streaming variant.

Training data and objective. During Audio LLM pretraining, we train the model on approximately 20 million hours of multilingual data constructed through our large-scale data pipeline, consisting primarily of ASR data and interleaved text–audio data. The pretrained audio encoder is kept frozen, while the audio adapter and the LLM decoder are jointly optimized. The model is trained with a cross-entropy loss over the target text tokens, enabling the audio adapter and LLM decoder to align acoustic representations with the LLM’s textual representation space. This stage establishes the fundamental audio–text alignment required for subsequent ASR adaptation.

The resulting Audio LLM combines the pretrained linguistic knowledge of Qwen with the audio–text alignment learned in this stage. It provides several useful priors for downstream ASR, including multilingual knowledge, world knowledge for resolving semantically ambiguous entities and homophones, and instruction-following capability for handling flexible textual conditions. Meanwhile, the MoE architecture preserves a large total model capacity while activating only part of parameters per token, maintaining relatively low decoding cost.

3.3 LLM Cooldown Adaptation

Following Audio LLM pretraining, we introduce a cooldown stage using approximately one million hours of higher-quality audio–text data. Compared with the preceding pretraining stage, this stage emphasizes a more balanced distribution of languages and Chinese dialects, improved transcription quality, and a broader mixture of tasks and instructions. In addition to conventional ASR, the training mixture includes audio understanding, speaker diarization, and speech question-answering and dialogue data.

Data rebalancing and quality filtering. We rebalance the multilingual and dialectal ASR data to improve coverage across linguistic conditions. To further improve transcription quality, we re-

transcribe the ASR data using VibeVoice and compare the resulting hypotheses with the existing reference transcripts. Samples with a word error rate greater than 0.3 are removed. This procedure uses agreement with an additional recognizer as a data-quality filter; it does not replace the reference transcripts with the re-recognition outputs.

Task diversification. The cooldown mixture broadens the training objectives beyond plain speech transcription. Audio understanding data introduce supervision for interpreting acoustic content, speaker diarization data expose the model to speaker-related structure, and speech question-answering and dialogue data enrich the range of audio-conditioned instructions and responses. Together with language and dialect rebalancing, this task mixture prepares the model for subsequent task-specific supervised fine-tuning.

Optimization. The pretrained SenseVoice audio encoder remains frozen, while the audio adapter and LLM decoder are optimized using a cross-entropy objective over the target output tokens. Relative to Audio LLM pretraining, the primary purpose of cooldown adaptation is to refine audio–language alignment using higher-quality, better-balanced data and to broaden task and instruction coverage before supervised fine-tuning.

3.4 Supervised Fine-tuning (SFT)

Supervised fine-tuning integrates the pretrained audio encoder, audio adapter, and cooldown-adapted LLM into an instruction-controlled audio-language model. Building on the recognition capabilities of the previous generation, the SFT mixture additionally incorporates speaker-diarized ASR and general audio understanding data. The task mixture is configured according to the capabilities of the corresponding model variant. During SFT, we jointly update the audio encoder and audio adapter, while keeping the base LLM parameters frozen and adapting the decoder through Low-Rank Adaptation (LoRA). Training examples follow a conversational format in which a user instruction and the associated audio are paired with an assistant response specifying the desired output. The model is optimized using an autoregressive cross-entropy objective over the target response. The ASR portion retains broad coverage of multilingual, dialectal, code-switching, noisy, far-field, and application-oriented speech. Recognition instructions may specify target languages, historical transcript context, hierarchical hotword candidates, or transcription polishing. The additional speaker-diarized ASR and audio understanding examples extend this instruction-following interface to structured transcription and audio-conditioned semantic tasks.

3.4.1 Recognition-oriented SFT

Specified-language transcription. To support flexible language control, the SFT mixture includes multilingual, dialectal, and code-switching speech covering the target languages and Chinese dialectal varieties. Each example may be accompanied by a list of target languages that specifies the expected transcription behavior. For monolingual transcription, the instruction requires the model to emit only speech belonging to the specified language and to avoid translating or rewriting speech from other languages into it. For multilingual speech, the model is instructed to preserve each supported language in its original written form. Training on diverse language combinations and code-switching examples enables the model to follow runtime language specifications and reduces unintended cross-language substitutions, particularly for lower-resource languages.

Domain-entity augmentation. To improve recognition of domain-specific entities that are individually rare in generic ASR data, we augment the SFT mixture with dedicated entity-focused examples. High-value entities are first mined from domains such as artificial intelligence, finance, healthcare, and IT, and an LLM generates diverse short utterances containing these entities based on retrieved factual context. The generated texts are converted into speech using CosyVoice3 [14], supplemented by human recordings for terms that are difficult to synthesize reliably. The resulting audio is then re-decoded by the current ASR model, and samples with unambiguous entity-level mismatches are filtered out. These curated examples are incorporated into SFT to increase the model’s exposure to rare and rapidly evolving domain terminology.

Native transcription polishing. A binary polishing instruction determines whether the target preserves the verbatim spoken form or produces a cleaned transcript. Polishing-enabled examples

teach the model to remove non-semantic fillers, collapse stuttering and unnecessary repetitions, resolve explicit self-corrections, and normalize punctuation and textual formatting while preserving the speaker’s final intent. The polishing data are constructed from both real and synthetic examples. For real speech, we retain the original audio from existing ASR records and regenerate the textual supervision using a text LLM with production polishing instructions, followed by fidelity filtering to remove samples with semantic loss or unsupported insertions. For synthetic augmentation, an LLM injects fillers, repetitions, and self-corrections into canonical transcripts, and the resulting texts are converted into speech using TTS, followed by ASR-based pronunciation filtering and noise augmentation. Since the polished transcript itself serves as the ASR target, recognition and polishing are learned within the same autoregressive decoding process, without requiring a cascaded ASR–LLM rewriting pipeline.

Historical-context conditioning. For contextual training, transcripts or interaction records from preceding turns are prepended as reference information. The instruction explicitly indicates that the historical context may be incomplete, irrelevant, or incorrect, and that the current audio should remain the primary source of evidence. The context data are constructed from both real conversations and targeted synthetic examples. For real data, we match ASR records to their original sessions using audio identifiers, restore the chronological order of utterances, and use only preceding turns as context without LLM rewriting. For synthetic augmentation, we construct examples involving homophone-ambiguous entities together with correct, conflicting, noisy, or empty historical contexts and hotword conditions. This mixture encourages the model to exploit recurring entities and topic cues when they are informative, while avoiding unconditional copying from the provided context.

Hierarchical hotword conditioning. To train hierarchical hotword conditioning, each utterance is paired with two tiers of candidate hotwords. **P0** contains high-confidence, high-priority terms, including key named entities and other contextually critical expressions, whereas **P1** contains a broader candidate pool. An LLM assigns candidates to the two tiers by semantically matching them against the reference transcription, promoting unambiguous contextual matches to P0 and retaining the remaining candidates in P1. During SFT, both candidate sets are included in the instruction as auxiliary conditioning inputs. The model learns to place greater preference on P0 candidates when supported by the acoustic and contextual evidence, while using P1 as a broader source of possible entities. This design improves long-tail hotword recall without forcing acoustically unsupported candidates into the transcript.

3.4.2 Speaker-diarized ASR

Training data. The training data combine carefully annotated human transcripts with model-generated supervision. For film and television data, we clean the recordings and use their original subtitles together with multiple specialized models, including language identification, sound event detection, and speaker diarization models, to cross-validate the annotations and curate high-quality speaker-attributed ASR examples. We additionally construct simulated meeting conversations with LLM-generated dialogue topics and varied reverberation and noise conditions to broaden coverage of conversational and acoustic scenarios.

Output serialization. Speaker-diarized ASR examples pair a multi-speaker recording with an instruction requesting both speaker attribution and transcription. The target response is serialized as newline-separated segments in the format `[SPK|start-end]text`. The example illustrates chronological serialization with overlapping time intervals and consistent speaker labels across turns.

Speaker-diarized transcription

[SPK0|19.36-23.36]

We can't merge until we know who owns the label definition for this release cycle.

[SPK2|25.20-30.00]

Well, we, we checked the documentation, yeah, and it is not clear at all.

[SPK3|26.08-26.48]

right.

[SPK0|28.32-33.60]

So the engineering estimate says it will take three days to fix the schema.

Segments are displayed in ascending order of start time; their time intervals may overlap. SPK3's brief response overlaps with SPK2, and SPK0 resumes before SPK2 finishes. Repeated speaker labels retain the same color. All timestamps are in seconds on an 80 ms grid.

Each segment contains a recording-local speaker label, a start–end time interval in seconds, and the corresponding recognized text. Repeated occurrences of the same speaker share the same label within the recording. The timestamps are expressed relative to the input audio, and intervals belonging to different speakers may overlap. The complete serialized response serves as the autoregressive training target, jointly supervising speaker labels, timestamps, and transcription text.

Speaker-token loss reweighting. To assess the balance of supervision across token categories, we examine token frequencies and loss contributions in the speaker-diarized ASR training data. Speaker-ID number tokens constitute 2.90% of all supervised tokens but contribute only 1.97% of the unweighted cross-entropy loss. Timestamp tokens account for 33.44% of tokens and 49.51% of loss, formatting tokens account for 14.48% and approximately 0.23%, and transcription tokens account for 46.50% and 47.74%, respectively. The remaining 2.68% of tokens and approximately 0.55% of loss are grouped as other supervised tokens, as shown in Figure 4.

These statistics reveal that speaker-ID prediction receives only a small share of the aggregate supervision despite its central role in speaker diarization. Timestamp prediction already accounts for a substantial fraction of the unweighted loss, while formatting is a relatively simple task that we observe the model already performs reliably. We therefore focus reweighting on speaker-ID prediction to better reflect its importance to diarization, rather than increasing the weight of every low-loss category. Specifically, we multiply the cross-entropy loss on speaker-ID number tokens by 10 while retaining unit weights for timestamp, formatting, transcription, and other supervised tokens. This adjusts the relative loss contributions without changing target sequences or token counts, and substantially improves both diarization error rate (DER) and concatenated minimum-permutation word error rate (cpWER) in our experiments.

Frame-aligned timestamp quantization. We also find that the temporal resolution of the supervision materially affects optimization. Our initial targets retained the original timestamps at 1 ms precision. We instead quantize timestamps to an 80 ms grid, matching the 12.5 Hz frame rate of the audio representations produced by the audio encoder and adapter (Figure 5). Compared with retaining the original 1 ms precision, this quantization yields substantially faster convergence of the training loss and of both DER and cpWER, as well as better DER and cpWER under the same training-data mixture. Together, speaker-token loss reweighting and frame-aligned timestamp quantization improve learning of the speaker and temporal structure represented in the serialized output.

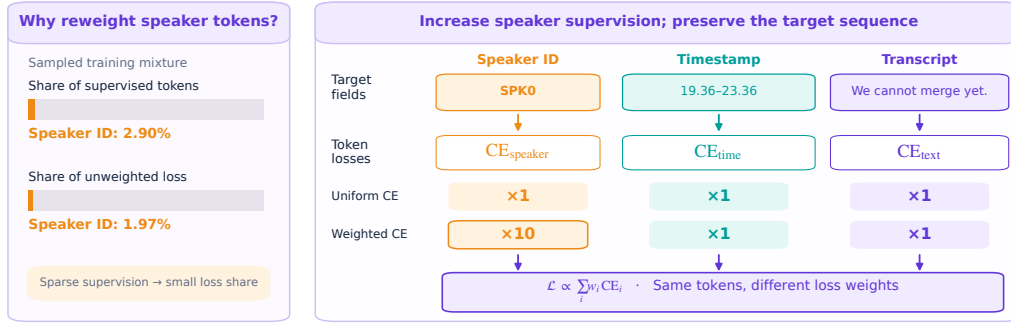


Figure 4: Speaker-token loss reweighting. Left: token-count and unweighted-loss shares for five categories in the speaker-diarized ASR training data. Other denotes supervised tokens outside the four named categories; its shares are calculated by subtraction. Formatting and other loss shares are approximate. Right: only speaker-ID number tokens receive a cross-entropy weight of 10; timestamp, formatting, transcription, and other supervised tokens retain a weight of 1. The example fields illustrate category membership rather than exact tokenizer segmentation. Target sequences and token counts remain unchanged.

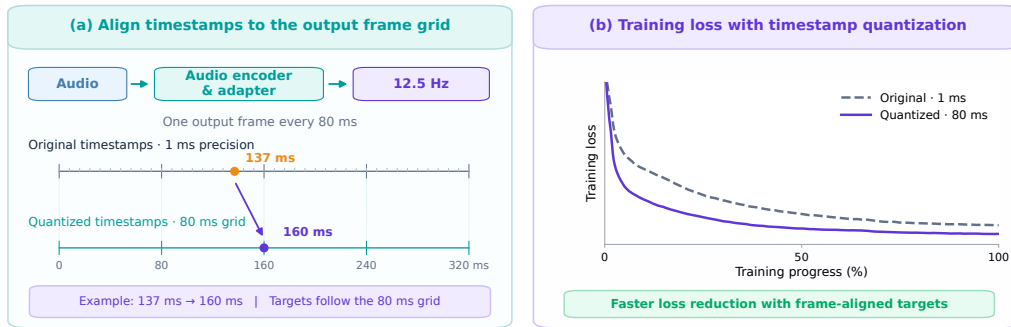


Figure 5: Frame-aligned timestamp quantization. (a) An illustrative timestamp is mapped from 1 ms precision to the 80 ms output frame grid of the audio encoder and adapter. (b) Training loss with original and quantized timestamps. Curves are digitized from training plots; absolute loss values and iteration counts are omitted. Both curves retain a shared vertical scale and use the same normalization of training progress.

3.4.3 General Audio Understanding

For the Next variant, SFT additionally includes audio understanding examples covering speech, environmental sounds, and music. These examples use task-specific instructions and reference responses within the same conversational training format.

Captioning examples supervise natural-language descriptions of audible content and acoustic characteristics. Temporal grounding examples pair an event description or query with the corresponding annotated time intervals, while sound event detection examples supervise the identification and temporal localization of target events. Audio question-answering examples provide questions about the recording and their reference answers, including both open-ended and multiple-choice formats.

All of these tasks are trained through autoregressive prediction of the corresponding target responses. The supervision therefore extends beyond lexical transcription to descriptions, temporal annotations, and answers grounded in the audio. Task instructions distinguish the requested output behavior, allowing the training mixture to cover different forms of audio understanding without treating every recording as a plain ASR example.

3.5 Reinforcement Learning (RL)

Following SFT, we apply reinforcement learning to further optimize Qwen-Audio-3.0-ASR for practical ASR objectives that are difficult to capture with maximum-likelihood training alone.

RL training data. We construct a compact, high-quality RL dataset focusing on challenging real-world cases. The dataset includes: (1) hard cases identified from disagreements between the base model and multiple external ASR systems, including Whisper, FireRed-ASR, and SenseVoice; (2) long-duration utterances exceeding 20 seconds; (3) hallucination-related samples, including both actual hallucinations and genuine long repetitions that may superficially resemble hallucinations; (4) keyword- and hotword-focused samples; and (5) regular ASR data included to mitigate catastrophic forgetting. Different sample types are associated with task-specific reward functions during RL, as described below.

The FunVerl-ASR Framework. We develop **FunVerl-ASR**, a fully asynchronous reinforcement learning framework purpose-built for large audio-language models (LALMs). Existing RL frameworks for LLMs, such as verl [15] and TRL [16], assume text-only inputs and do not natively support the audio encoder that Qwen-Audio-3.0-ASR requires to convert speech waveforms into token-aligned embeddings. FunVerl-ASR addresses this gap by extending the verl infrastructure with audio-aware data pipelines, a dedicated audio feature extraction stage, and an asynchronous training loop that decouples rollout generation from policy optimization.

As illustrated in Figure 6, FunVerl-ASR decomposes RL training into four concurrent modules orchestrated by Ray:

- **Audio Processor.** Raw audio waveforms are converted into Mel-spectrogram features by the audio processor, while instruction text (including context, hotwords, and other flexible prompts) is tokenized by the Qwen tokenizer. Both outputs are passed together to the rollout engine.
- **vLLM Rollout Engine.** The vLLM-based inference engine receives Mel features and token IDs jointly. Inside the engine, the frozen audio encoder and adapter transform Mel features into frame-level audio embeddings, which are then merged into the text embedding sequence. The combined embedding sequence is decoded by the MoE LLM to generate G candidate hypotheses per prompt.
- **Reward Module.** Each generated hypothesis is evaluated by a configurable reward module that supports rule-based scoring (CER, keyword recall, hotword hit, context entity match, hallucination detection, language consistency), LLM-judge scoring, and per-sample reward routing for mixed-domain batches.
- **Megatron Policy Trainer.** The actor model is trained using Megatron-Core with combined tensor, pipeline, and expert parallelism. After each gradient update, an NCCL-based checkpoint engine synchronizes the updated LLM decoder weights to the vLLM rollout workers, maintaining near-on-policy training without requiring full model reloading. The audio encoder remains frozen throughout RL and is not included in the parameter synchronization.

The key architectural innovation of FunVerl-ASR is its fully asynchronous training loop: the rollout engine continuously generates trajectories and pushes them into a message queue, while the policy trainer consumes batches from the queue and performs gradient updates independently. This eliminates the synchronous barrier between generation and training that dominates wall-clock time in conventional RL frameworks. A staleness threshold ensures that trajectories generated under excessively outdated policies are discarded, bounding the off-policy gap. The default configuration generates 8 hypotheses per prompt with a mini-batch size of 32 prompts per actor update.

We evaluate FunVerl-ASR’s training efficiency on a 4-node cluster (32 A100 GPUs, split equally between rollout and training). The entire RL training of Qwen-Audio-3.0-ASR completes within one day using this configuration.

GRPO-based RL for ASR. Based on the FunVerl-ASR framework, we employ Group Relative Policy Optimization (GRPO) [17] as the RL algorithm. GRPO is a lightweight policy-based method that generates a group of responses $\{o_i\}_{i=1}^G$ per prompt and computes rule-based rewards $\{R_i\}_{i=1}^G$ without requiring a separately trained value network. The group-level rewards are normalized to compute the advantage:

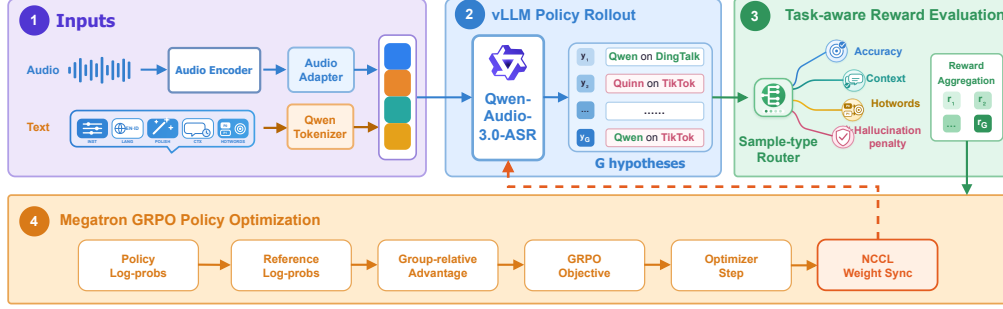


Figure 6: Overview of the FunVerl-ASR fully asynchronous RL framework for Qwen-Audio-3.0-ASR. The audio processor extracts Mel features and tokenizes instruction text; the vLLM rollout engine runs the frozen audio encoder internally and generates hypotheses from merged audio-text embeddings; the reward module scores each hypothesis; and the Megatron policy trainer performs GRPO updates with NCCL-based weight synchronization back to the rollout engine.

$$\hat{A}_{i,t} = \frac{R_i - \text{mean}(\{R_j\}_{j=1}^G)}{\text{std}(\{R_j\}_{j=1}^G)} \quad (3)$$

The policy is optimized with a clipped objective and a KL penalty term:

$$L_{\text{GRPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_{i,t}) - \beta D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \quad (4)$$

where $r_{i,t}(\theta) = \pi_{\theta}(o_{i,t} \mid q, o_{i,<t}) / \pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})$.

When WER is used as the reward, GRPO is reduced to a form closely related to Minimum Word Error Rate (MWER) training, a well-established criterion in the ASR community. Beyond basic WER, we design a composite reward function with per-sample routing to handle mixed-domain training batches:

- **ASR Accuracy** (R_i^1). $1 - \text{WER}(y^*, y)$ as the primary reward, with range $[0, 1]$.
- **Keyword Accuracy and Recall** (R_i^2). Keyword recall encourages correct entity recognition; keyword accuracy penalizes insertion errors to balance precision and recall. Keywords are either manually annotated or LLM-identified.
- **Hotword Reward** (R_i^3). For hotword-conditioned samples, a dedicated reward scores recognition of P0/P1 hotwords provided in the prompt, with asymmetric positive/negative gamma weighting.
- **Context Reward** (R_i^4). For context-conditioned samples, rewards hit/miss of context-derived entities with configurable penalties for false insertions.
- **Hallucination Suppression** (R_i^5). Hallucinated content is detected via pattern matching and penalized proportionally to the hallucinated span length.

FunVerl-ASR’s reward routing mechanism selects the appropriate reward configuration per sample based on its data source (general, dialect, hotword, or context), enabling a single training run to jointly optimize across all production scenarios.

4 Deployment

Qwen-Audio-3.1-ASR provides deployment variants for file transcription, low-latency streaming, context-aware voice input, speaker-diarized transcription, and general audio understanding. The Streaming and Message ASR profiles retain the workflows of Qwen-Audio-3.0-ASR. The family additionally introduces explicit selection between dialect transcription and dialect-to-Mandarin translation,

as well as speaker-diarized transcription and general audio understanding in the corresponding variants. These capabilities are variant-dependent rather than uniformly available across all deployment profiles.

4.1 Multilingual and Dialectal ASR

Qwen-Audio-3.1-ASR retains the instruction-controlled multilingual recognition interface of Qwen-Audio-3.0-ASR, supporting 30 languages and 16 Chinese dialectal varieties spanning eight major dialect regions. Applications can specify target languages at inference time to accommodate multilingual and code-switching speech without switching between language-specific checkpoints. For multilingual transcription, supported languages are preserved in their original writing systems.

A new capability in this generation is user-selectable output behavior for Chinese dialectal speech. Through an explicit task instruction, users can choose between ASR and automatic speech translation (AST):

- **Dialect ASR.** The model transcribes the dialectal speech while preserving its dialect-specific wording and expressions, rather than translating them into Mandarin.
- **Dialect-to-Mandarin AST.** The model translates the dialectal speech into written Mandarin, conveying the meaning of the utterance rather than preserving its original dialectal expressions.

This explicit choice separates faithful dialect transcription from dialect-to-Mandarin translation. It allows applications to select the appropriate output for scenarios such as dialect documentation or cross-dialect communication. The ASR/AST selection specifies the linguistic task and is distinct from transcription polishing, which controls the removal of disfluencies and the presentation of recognized text.

4.2 Low-Latency Streaming Recognition

The streaming deployment profile retains the hybrid incremental and full-context recognition workflow of Qwen-Audio-3.0-ASR. Incoming audio is processed incrementally to provide provisional transcripts before an utterance is complete. After endpoint detection, the complete utterance is used to refresh and finalize the transcript. This workflow combines immediate textual feedback with recognition informed by the full utterance context.

Low-latency recognition is achieved by processing audio in short chunks with limited right context. Chunk size and right-context length provide configurable operating points for the latency–accuracy trade-off: shorter contexts prioritize responsiveness, while longer contexts provide additional acoustic evidence before the transcript is updated. Partial hypotheses may be revised as more audio becomes available.

The streaming variant supports incremental speech recognition but does not provide speaker-diarized transcription or general audio understanding. These capabilities are available through the corresponding non-streaming variants.

4.3 Message ASR for Context-Aware Transcription

The Message ASR deployment profile follows Qwen-Audio-3.0-ASR, retaining its contextual conditioning, hierarchical hotword customization, and native transcription polishing workflow. It is designed for applications such as voice input and conversation documentation, where spontaneous speech must be converted into readable text and terminology may recur across successive utterances.

Historical conversation turns or preceding transcript segments provide topic and entity cues. Hierarchical hotword candidates distinguish a small set of high-confidence terms from a broader candidate pool. These conditions guide recognition together with the acoustic input, supporting the transcription of names, products, and specialized terminology.

A polishing instruction controls whether the output preserves the spoken form or produces a cleaned transcript. When enabled, native polishing removes non-semantic fillers and redundant repetitions, resolves explicit self-corrections, and normalizes punctuation and formatting while preserving the speaker’s intended meaning. This processing takes place within recognition rather than through a separate post-recognition rewriting service.

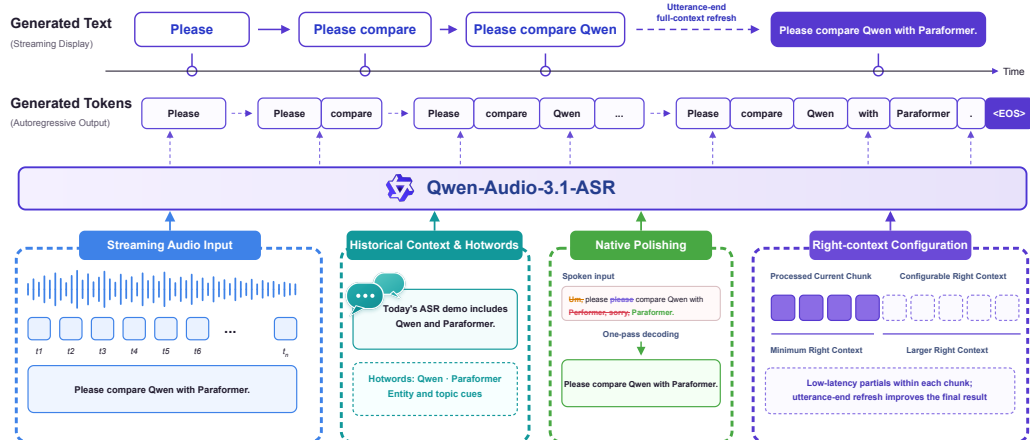


Figure 7: Overview of the Message ASR workflow used in Qwen-Audio-3.1-ASR, following the design of Qwen-Audio-3.0-ASR. Streaming audio is decoded incrementally to provide responsive partial transcripts. Historical context supplies entity and topic cues, native polishing improves readability within the same decoding pass, and configurable right context balances streaming latency and recognition accuracy. At utterance end, the model refreshes the displayed hypothesis using the complete available context.

In the interactive workflow, incremental hypotheses provide immediate feedback, and an utterance-end full-context refresh produces the finalized transcript. Language specifications, historical context, hotwords, and polishing can be configured according to the application. These controls remain unchanged from the previous generation and do not imply support for speaker-diarized output or general audio understanding in the Message ASR profile.

4.4 Speaker-Diarized Transcription

The non-streaming Flash and Next variants support speaker-diarized transcription for multi-speaker recordings. Given an audio input and a corresponding task instruction, the model returns recognized text associated with recording-local speaker labels and temporal boundaries.

The output uses the segment format [SPK|start-end]text, where the time interval is expressed in seconds relative to the input recording. Repeated occurrences of a speaker retain the same label within the recording, and intervals from different speakers may overlap. This representation preserves speaker turns, brief responses, and simultaneous speech rather than reducing the conversation to a single unstructured transcript.

The resulting segments support speaker-aware transcript display, navigation to the corresponding audio intervals, and downstream analysis of conversational content. Speaker labels distinguish participants within a recording and do not imply real-world identity recognition.

4.5 General Audio Understanding

The Next variant supports instruction-conditioned understanding of speech, environmental sounds, and music. Applications provide an audio recording together with a task instruction or question, and the model produces the corresponding textual response or temporal annotations.

Captioning. Applications can request natural-language descriptions of audible content, including acoustic events, spoken delivery, and musical characteristics. This mode provides information beyond verbatim transcription and applies to recordings with or without speech.

Temporal grounding and sound event detection. A query can specify acoustic content to be localized within the recording. The output identifies the relevant time intervals, allowing applications to retrieve or inspect the corresponding audio segments. Sound event detection similarly supports locating occurrences of requested events, including multiple occurrences within the same recording.

Audio question answering. Applications can ask open-ended questions or provide multiple-choice questions about the audio. Questions may concern audible content, acoustic characteristics, or temporal relationships among events. This interface supports audio-conditioned analysis without first reducing the recording to a textual transcript.

These modes complement speech recognition rather than replace it. ASR provides the spoken text, speaker-diarized transcription adds speaker and temporal structure, and general audio understanding addresses tasks requiring broader acoustic evidence.

5 Evaluations

We evaluate Qwen-Audio-3.0-ASR on both open-source ASR benchmark datasets and real-world industry evaluation sets. For the open-source evaluation, we use corresponding test sets of AISHELL-1 [18], AISHELL-2 [19], FLEURS [20], LibriSpeech [21], WeNetSpeech [22], GigaSpeechBench [23] and CommonVoice [24] data sets. In addition to these public benchmarks, we evaluate the system on a broad set of internal industrial test sets designed to reflect practical deployment conditions. These evaluations cover industry-specific entities and long-tail terminology, hotword customization, long-audio contextual recognition, transcription polishing, and low-latency streaming recognition. The internal test sets span diverse acoustic conditions, speaking styles, application domains, and usage scenarios, providing a complementary assessment of the system’s production-oriented capabilities.

5.1 Public Benchmark Results

5.1.1 Chinese and English ASR Benchmarks

We evaluate Qwen-Audio-3.0-ASR against a range of publicly released models and commercial ASR systems on widely used Chinese and English benchmarks, as shown in Table 1. Panel A compares our model with results published in official model repositories or technical reports, whereas Panel B compares systems evaluated using our unified API-based evaluation pipeline. We therefore discuss the two panels separately.

Comparison with officially reported results. As shown in Panel A, Qwen-Audio-3.0-ASR has the lowest reported error rate on four of the seven test sets. Compared with the strongest officially reported baseline on each of these test sets, its error rate is lower by 0.08 percentage points (3.8% relative) on AISHELL-2, 0.60 points (22.4% relative) on FLEURS-zh, 0.18 points (7.4% relative) on LibriSpeech-other, and 0.36 points (7.7% relative) on WeNetSpeech-net. It also remains within 0.02 percentage points of the best officially reported result on LibriSpeech-clean. Because these baseline results are collected from different public sources, they may reflect source-specific preprocessing, reference transcripts, and scoring conventions. These numerical differences should therefore be interpreted with caution, particularly for WeNetSpeech-net, where our evaluations use corrected reference transcripts.

Comparison under our unified evaluation pipeline. As shown in Panel B, Qwen-Audio-3.0-ASR achieves the lowest error rate on four of the seven test sets. Relative to the strongest competing system evaluated using the same pipeline, it reduces the error rate by 0.30 percentage points (12.6% relative) on FLEURS-zh, 0.78 points (15.8% relative) on FLEURS-en, 0.16 points (11.9% relative) on LibriSpeech-clean, and 0.27 points (10.8% relative) on LibriSpeech-other. It additionally remains within 0.09 percentage points of the best result on AISHELL-2. These results demonstrate competitive recognition performance across both Chinese and English benchmarks.

5.1.2 Multilingual ASR Benchmarks

To further assess the multilingual capabilities of the system, we evaluate Qwen-Audio-3.0-ASR on three public multilingual benchmarks, including GigaSpeechBench, Common Voice 15, and FLEURS, using WER or CER according to the language-specific evaluation protocol. As shown in Table 2, GigaSpeechBench covers seven East and Southeast Asian languages, on which Qwen-Audio-3.0-ASR achieves a macro-average error rate of 22.50%, the lowest among systems evaluated on all seven languages, and records the lowest error rate on five of them. On Common Voice 15,

Table 1: Evaluation results on public ASR benchmark test sets. We report CER (%) for Chinese test sets and WER (%) for English test sets. For the competing systems, Panel A presents results taken from official public repositories or technical reports, whereas Panel B presents results obtained using our unified API-based evaluation pipeline on the same test sets. Qwen-Audio-3.1-ASR is repeated in both panels for reference. Lower values indicate better performance, and the best result within each panel on each test set is shown in bold. Because the results in Panel A may involve source-specific preprocessing and scoring conventions, direct quantitative comparisons are primarily made within Panel B. For WeNetSpeech-net, all results in Panel B and our model’s results in Panel A are evaluated using corrected reference transcripts, following the audio–label mismatch discussion in [WenetSpeech issue #63](#). The competing-system results in Panel A are retained as reported in their original sources.

Panel A: Officially Reported Results								
Test set	Metric	Whisper -large-v3	Kimi -Audio	Step -Audio 2	FireRed -ASR	LongCat-Flash- Omni Instruct	Qwen-Audio -3.1-ASR	
AISHELL-1	CER	4.72	0.71	0.63	0.54	0.63	1.08	
AISHELL-2	CER	4.68	2.86	2.10	2.58	2.78	2.05	
FLEURS-zh	CER	5.18	3.11	2.68	4.81	3.99	2.25	
FLEURS-en	WER	6.23	6.99	3.03	10.79	5.02	4.72	
LibriSpeech-clean	WER	1.86	1.32	1.17	1.84	1.57	1.23	
LibriSpeech-other	WER	3.43	2.63	2.42	4.52	4.01	2.19	
WeNetSpeech-net	CER	11.89	6.45	4.67	4.94	6.09	4.37	
Panel B: Results from Our API Evaluation								
Test set	Metric	GPT-4o Transcribe	Azure MAI- Transcribe-2	Doubao-ASR	Fun-ASR -Flash	Tencent Hy-ASR-3.0- preview	Muse-Voice Transcribe-1.0	Qwen-Audio -3.1-ASR
AISHELL-1	CER	3.34	1.49	1.54	0.76	0.78	6.28	1.08
AISHELL-2	CER	4.47	2.82	2.66	1.93	2.15	6.80	2.05
FLEURS-zh	CER	6.14	5.38	8.18	5.52	2.38	8.34	2.25
FLEURS-en	WER	5.79	5.38	6.50	4.95	5.45	7.48	4.72
LibriSpeech-clean	WER	1.58	1.68	3.36	1.35	1.60	2.17	1.23
LibriSpeech-other	WER	3.87	3.13	4.46	2.51	2.68	5.10	2.19
WeNetSpeech-net	CER	12.27	4.38	4.81	3.83	4.16	10.84	4.37

which covers five Asian languages together with Spanish and French, the model achieves the lowest macro-average error rate of 4.57% and the lowest error rate on six of the seven languages. On FLEURS, which evaluates seven Asian languages together with Spanish and Portuguese, the results are more mixed: Qwen-Audio-3.0-ASR records the lowest error rate on four of the nine languages, with a macro-average error rate of 4.94%.

5.2 Chinese Dialect ASR and AST

We evaluate Qwen-Audio-3.1-ASR against Doubao-ASR and Tencent Hy-ASR-3.0-preview on internal test sets and public dialect benchmarks. The internal evaluation includes an ASR configuration covering 16 Chinese dialects and an AST configuration covering 11 dialects. ASR requires transcription in the corresponding dialect, whereas AST requires translation of dialectal speech into Mandarin. These configurations are evaluated separately rather than assigning each dialect exclusively to one task.

For the internal ASR evaluation, we report character error rate (CER). For the internal AST evaluation, we report semantic sentence accuracy assessed by Qwen3.7-Max, computed as the complement of the LLM-assessed P0 error rate:

$$\text{Semantic Sentence Accuracy}(\%) = 100(1 - p_{P0}), \quad (5)$$

where $p_{P0} \in [0, 1]$ is the proportion of evaluated utterances classified as P0 semantic errors. Thus, when the P0 rate is expressed as a percentage, semantic sentence accuracy is obtained by subtracting it from 100. CER measures transcription errors, while semantic sentence accuracy measures sentence-level semantic correctness in the AST outputs. Macro averages are computed with equal weight for each included subset, separately for each evaluation configuration.

Table 2: Error rates (%) of multilingual ASR systems on Common Voice 15, FLEURS, and GigaSpeechBench. The evaluation metric for each language is given in the second column. Japanese CER for the previously reported systems follows the normalization and scoring protocol specified in [Fun-ASR PR #165](#). Macro averages are calculated only for systems with results on all languages included in the corresponding benchmark. Bold values denote the lowest error rate among systems evaluated on the corresponding language.

Language	Metric	Doubao-ASR	Tencent Cloud ASR	Gemini 3.1 Pro	GPT-4o Transcribe	Azure MAI-Transcribe-2	Muse Voice Transcribe-1.0	Qwen-Audio -3.1-ASR
GigaSpeechBench								
Indonesian	WER	27.73	32.55	24.47	39.52	29.18	22.90	16.69
Japanese	CER	29.47	34.56	39.12	45.46	34.17	32.05	18.47
Thai	CER	17.65	31.40	17.37	53.55	22.42	15.27	7.75
Filipino	WER	25.36	33.38	37.41	37.73	32.95	25.48	21.69
Vietnamese	WER	9.53	9.61	15.55	32.89	12.77	10.76	8.04
Korean	CER	14.52	15.00	19.70	32.52	15.96	16.52	7.98
Malay	WER	35.05	41.06	40.19	51.59	38.18	38.81	25.04
Macro average	–	22.76	28.22	27.69	41.89	26.52	23.11	15.09
Common Voice 15								
Indonesian	WER	4.25	7.79	12.92	8.74	4.33	5.98	3.32
Japanese	CER	6.44	12.02	11.52	11.22	7.21	12.24	1.40
Thai	CER	2.32	4.17	12.33	6.64	1.90	2.94	1.43
Vietnamese	WER	6.31	5.67	10.64	15.50	5.93	7.77	8.44
Korean	CER	3.89	8.16	4.86	5.80	2.89	7.49	0.73
Spanish	WER	5.42	4.95	3.97	4.18	3.64	5.09	3.23
Portuguese	WER	8.06	6.24	8.65	8.78	4.89	6.49	4.47
French	WER	8.89	22.06	10.51	9.26	7.38	10.72	6.87
Macro average	–	5.70	8.88	9.43	8.77	4.77	7.34	3.74
FLEURS								
Indonesian	WER	5.24	12.77	3.72	3.73	2.62	8.16	2.01
Japanese	CER	3.16	15.39	3.95	2.98	1.85	6.55	0.36
Thai	CER	6.18	11.38	6.05	4.99	3.85	8.27	5.37
Filipino	WER	10.34	18.05	8.33	7.42	5.98	9.95	10.80
Vietnamese	WER	6.32	15.19	4.70	3.23	2.42	3.97	4.46
Korean	CER	4.86	8.93	4.24	4.06	3.92	6.87	2.38
Malay	WER	8.14	17.13	4.22	4.21	4.08	9.54	5.12
Spanish	WER	5.31	8.09	2.62	2.13	2.04	5.61	1.95
Portuguese	WER	7.19	13.71	3.46	2.80	2.53	7.12	2.21
Macro average	–	6.30	13.40	4.59	3.95	3.25	7.34	3.85

As shown in Figure 8, Qwen-Audio-3.1-ASR achieves the lowest CER on 11 of the 16 internal ASR subsets. Its macro-average CER is 10.38%, compared with 20.23% for Doubao-ASR and 17.13% for Tencent Hy-ASR-3.0-preview, corresponding to absolute reductions of 9.85 and 6.75 percentage points, respectively. The largest improvement occurs on Wenzhou, where CER decreases to 19.21%, compared with 68.22% and 64.74% for the two baselines. Qwen-Audio-3.1-ASR also achieves the lowest CER on Shanghai, Nanchang, Sichuan, Ningbo, Hakka, Hangzhou, Henan, Hunan, Fujian, and Suzhou.

On the internal AST test set, Qwen-Audio-3.1-ASR achieves the highest semantic sentence accuracy on 10 of the 11 subsets (Figure 9). Its macro-average accuracy is 82.10%, compared with 57.75% for Doubao-ASR and 58.99% for Tencent Hy-ASR-3.0-preview, yielding gains of 24.35 and 23.11 percentage points, respectively. On Wenzhou, accuracy reaches 71.00%, compared with 2.38% and 4.33% for the baselines. Substantial gains are also observed on Ningbo, Hangzhou, and Suzhou. Cantonese is the exception: Qwen-Audio-3.1-ASR obtains 93.55%, slightly below Tencent Hy-ASR-3.0-preview at 94.15%.

We additionally evaluate CER on 11 public benchmark subsets: nine KeSpeech subsets under the AST setting and the long- and short-form WSYue Cantonese subsets under the ASR setting. The four WSC Sichuan subsets are excluded. As shown in Figure 10, Qwen-Audio-3.1-ASR achieves a macro-average CER of 4.55%, compared with 7.15% for Doubao-ASR and 4.61% for Tencent Hy-ASR-3.0-preview. It achieves the lowest CER on six subsets: KeSpeech-Beijing, KeSpeech-Ji-Lu, KeSpeech-Jiang-Huai, KeSpeech-Jiao-Liao, KeSpeech-Mandarin, and WSYue-long. These results

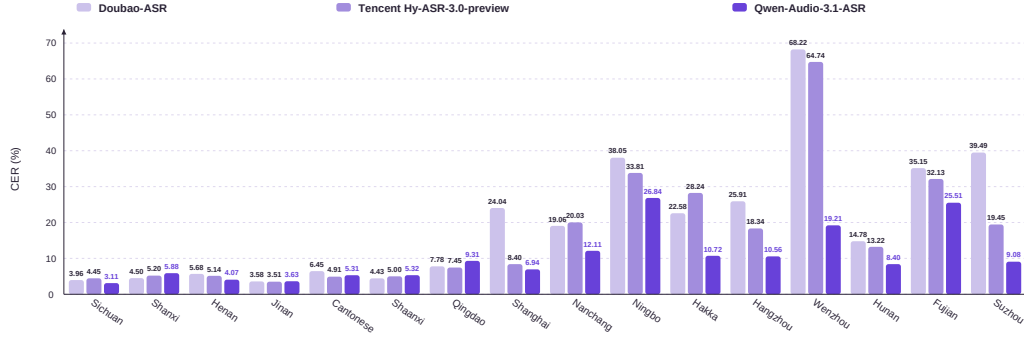


Figure 8: Character error rate (CER, %) on the internal ASR test set covering 16 Chinese dialects. All subsets use the ASR setting. Lower is better.

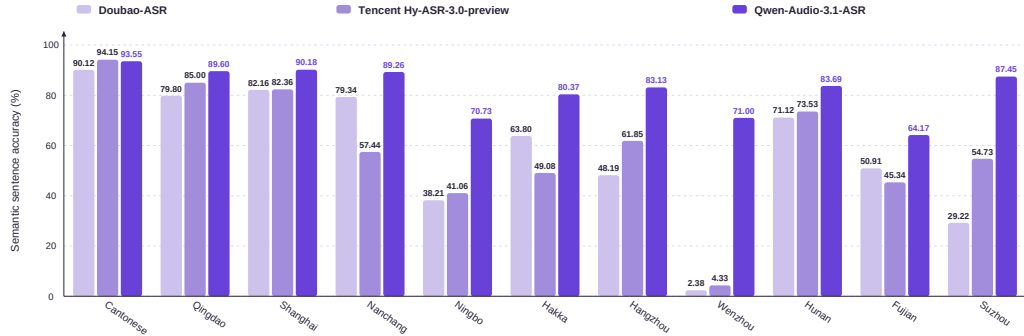


Figure 9: Semantic sentence accuracy (%) on the internal AST test set covering 11 Chinese dialects. Outputs are Mandarin translations of dialectal speech. Accuracy is computed as the complement of the LLM-assessed P0 error rate, using Qwen3.7-Max as the evaluator. Higher is better.

indicate a substantial improvement over Doubao-ASR and a small macro-average advantage over Tencent Hy-ASR-3.0-preview on the selected public benchmarks.

5.3 Industry-domain Entity Recognition

Beyond general-purpose recognition accuracy, real-world ASR systems require reliable recognition of domain-specific terminology, including financial terms, medical terminology, software and IT vocabulary, organization names, and other long-tail entities that occur relatively infrequently in generic training data and are therefore prone to substitution or omission errors. To improve recognition of such entities, we continuously mine domain-specific entity vocabularies and construct corresponding ASR training data through the entity-mining and data-synthesis pipeline described in Section 3.4.

Figure 11 reports entity recall on an internal industry-domain evaluation set covering 15 domains, comparing Doubao-ASR, Tencent Hy-ASR-3.0-preview, and Qwen-Audio-3.0-ASR. Qwen-Audio-3.0-ASR achieves the highest entity recall across all evaluated domains. The improvements are particularly pronounced in Software & IT, where the evaluation contains many low-frequency and rapidly evolving technical entities such as software libraries, frameworks, and API names. Consistent improvements are also observed across domains including Artificial Intelligence, Healthcare, Agriculture, Manufacturing, and Culture, Sports & Media. Overall, these results demonstrate the effectiveness of Qwen-Audio-3.0-ASR in recognizing long-tail industry-specific entities across a diverse range of domains. The observed gains are also consistent with the expected benefits of the entity-mining and data-synthesis pipeline, although other improvements introduced in Qwen-Audio-3.0-ASR may also contribute to the overall performance gains.

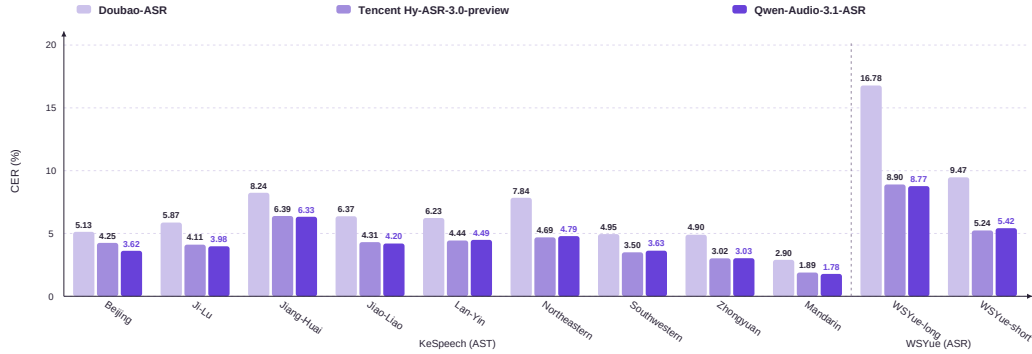


Figure 10: CER (%) on public dialect ASR and AST benchmarks, comprising nine KeSpeech AST subsets and two WSYue Cantonese ASR subsets. The four WSC Sichuan subsets are excluded from this comparison and its macro averages. Lower is better.

5.4 Hotword Customization

Within Qwen-Audio-3.0-ASR, we introduce a hierarchical hotword customization mechanism. Retrieved hotword candidates are divided into two priority levels: a high-priority set (**P0**) and a broader candidate set (**P1**). During decoding, the two sets are assigned different conditioning strengths, with P0 receiving a larger weight than P1. This design allows the model to emphasize high-confidence hotwords while still benefiting from a broader set of candidate terms.

Table 3 reports hotword recall with and without hotword conditioning across several long-tail entity categories. With hotword conditioning enabled, Qwen-Audio-3.0-ASR achieves the highest recall on all evaluated P0 categories, demonstrating the effectiveness of the hierarchical hotword mechanism for improving recognition of high-priority long-tail entities.

For subject terms and trending buzzwords, the improvement in P1 recall is smaller than that observed for the corresponding P0 categories, which is consistent with the weaker conditioning weight assigned to P1 candidates.

Table 3: Hotword recall (%) without vs. with hotword conditioning, by category and priority tier (P0/P1).

Category	Doubao-ASR		Fun-ASR-Flash		Qwen-Audio-3.0-ASR	
	No hotword	+Hotword	No hotword	+Hotword	No hotword	+Hotword
Person names (P0)	32.27	82.64	37.33	95.26	62.12	99.43
Subject terms (P0)	75.14	93.06	71.84	92.53	69.36	99.42
Subject terms (P1)	84.36	93.04	86.02	96.73	88.46	95.31
Trending buzzwords (P0)	48.24	91.95	43.26	82.53	63.08	99.46
Trending buzzwords (P1)	51.24	89.48	53.92	87.00	74.00	88.34
AI entities (P0)	44.21	65.24	79.88	90.55	88.22	99.39
Brand names (P0)	44.18	57.88	80.14	95.55	83.83	97.95
Product entities (P0)	67.59	78.70	73.64	89.92	72.87	99.07

5.5 Long-audio Context Modeling

Many everyday recognition errors are homophone or first-mention-entity confusions that are only resolvable with context beyond the current utterance – for example, disambiguating a technical filename from a similar-sounding common phrase, or recognizing a person’s name correctly only after it has already appeared earlier in the same conversation. Frame-local acoustic decoding cannot resolve these cases because its receptive field does not extend across utterance boundaries.

Building on the contextual supervised fine-tuning described in Section 3.4, Qwen-Audio-3.0-ASR conditions the transcription of the current audio segment on recently recognized text from the same session, functioning as a running context cache that requires no manually maintained hotword list.

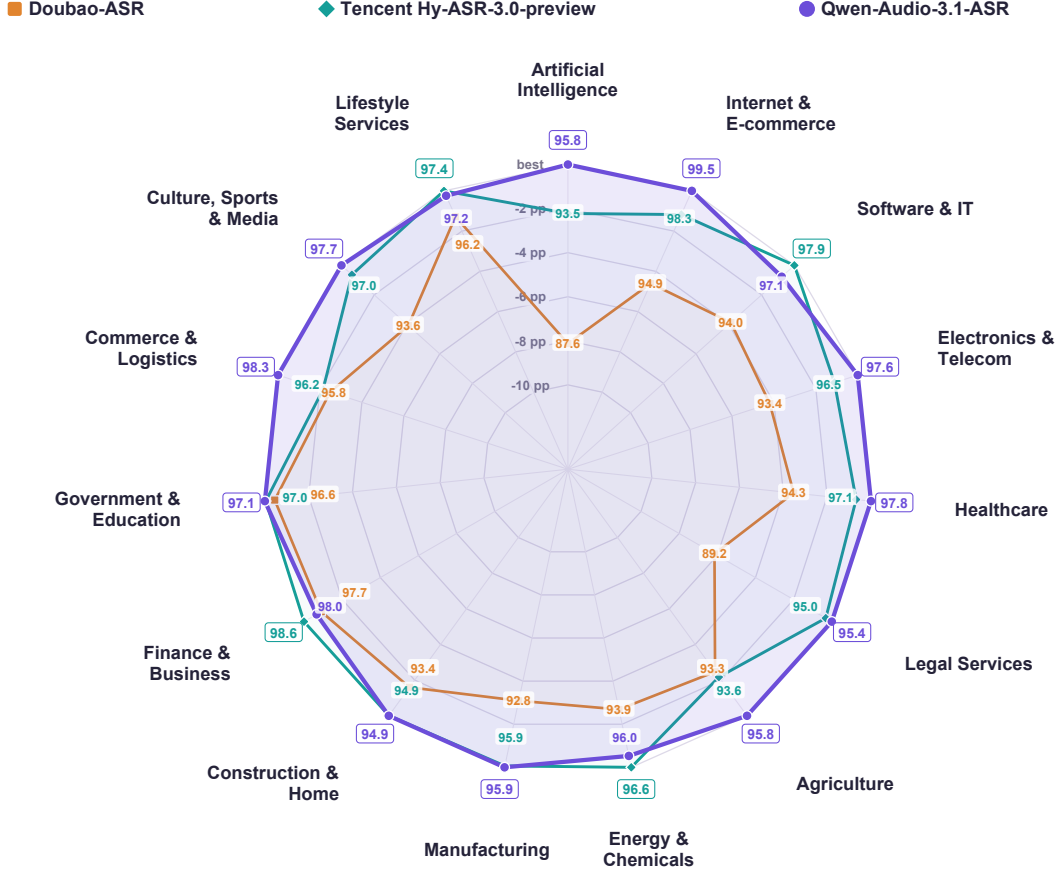


Figure 11: Industry-domain entity recall across 15 domains. For visualization, radial positions are normalized independently within each domain so that the best-performing system lies on the outer ring; numerical labels report the original entity-recall values (%).

Table 4 illustrates representative corrections enabled by this mechanism: entities and technical terms that recur across segments (e.g., a codebase name, a repeated technical abbreviation, or a person’s name introduced earlier) are transcribed consistently once they are anchored by prior context, whereas the same segment decoded without that context reverts to a phonetically plausible but incorrect alternative.

5.6 Native Transcription Polishing

Conventional pipelines address filler words, stutter-like repetitions, and mid-sentence self-corrections through a second, cascaded stage: an LLM rewrites the raw ASR transcript after it is produced. This decouples acoustic decoding from text refinement, but it adds a second autoregressive generation pass whose latency lands directly on the critical path between end-of-utterance and text availability.

Qwen-Audio-3.0-ASR instead performs this cleanup within the same decoding pass as recognition: an instruction flag toggles whether the model emits the raw transcript or a polished version that removes filler words and stutter-like repetitions, resolves self-corrections in favor of the speaker’s final intent, and normalizes punctuation and formatting, all without an additional model call. Figure 12 compares this native, single-pass polishing against the same model’s raw output and against a conventional two-call cascade (the same raw ASR output rewritten by Qwen3.6-Plus). Native polishing raises the readability score from 2.53 to 3.44 (on an internal 5-point scale), reaching a faithfulness score (3.44) matching the two-call cascade closely (3.47), while requiring only a single inference pass instead of two.

Table 4: Representative corrections enabled by long-audio context. The context cues occurred earlier in the same session. Colored spans highlight recognition errors and their context-conditioned corrections.

Earlier context cue	Reference	Without long context	With long context
xsltproc style.xml input.xml occurred earlier	style.xml data.xml > out.html	xsl data.xml > out.html	style.xml data.xml > out.html
Earlier explanation: “这叫法条竞合”	法条竞合	划掉即可	法条竞合
The term unigram occurred twice earlier	我们称之为unigram	我们称之为unique	我们称之为unigram
TCC was repeatedly established as the metric name	our TCC will be...	our GGC will be...	our TCC will be...
The name “洛军是阿占的儿子” occurred earlier	洛军是阿占的儿子	洛君是阿占的儿子	洛军是阿占的儿子
The technical term softmax occurred earlier	softmax 形式	softmap 形式	softmax 形式
The mode name debug occurred earlier	只能debug 模式	只能第八个 模式	只能debug 模式
The name “张贵生” occurred earlier	张贵生	张桂生	张贵生

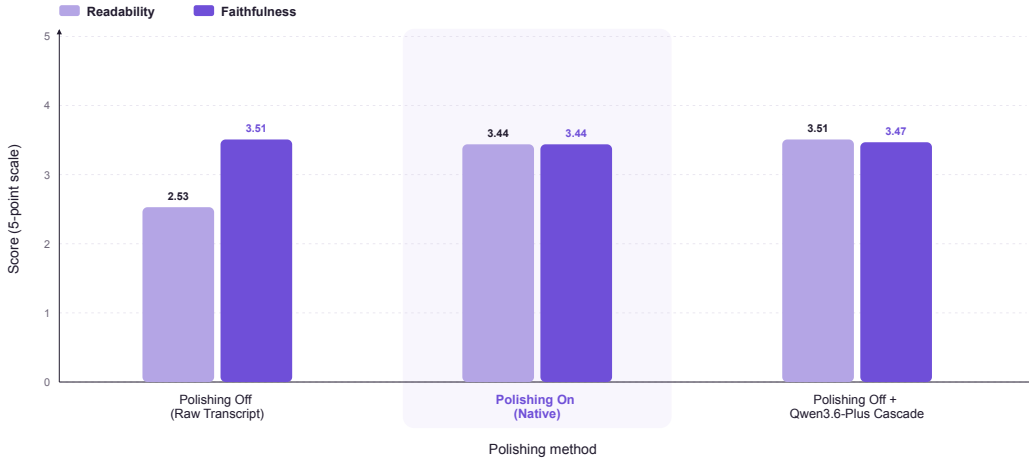


Figure 12: Readability and faithfulness of native polishing vs. the raw transcript and vs. an external ASR+LLM cascade.

Figure 13 presents representative Chinese and English polishing examples. Taking the English cases as an illustration, the model removes hesitation fillers such as *um* and *uh*, collapses repetitions such as *the the the*, and resolves false starts by retaining the speaker’s final intended expression. For example, the fragmented phrase *I just to put it / some comparison* is consolidated into *Just to put it in some comparative context*. The model also normalizes capitalization, punctuation, and acronyms, converting the spoken letter sequence *d r r* into *DRR*. These examples show that native polishing produces more fluent and readable transcripts while largely preserving the speaker’s original meaning.

5.7 Low-Latency Streaming Recognition

Streaming recognition performance is evaluated on internal industrial test sets for Chinese and English. We evaluate multiple streaming configurations with different audio chunk sizes and right-context windows, resulting in the algorithmic-latency operating points shown in Figure 11. All online decoding and latency measurements are performed on an NVIDIA H100 GPU. Both first-token latency and text-display latency are measured from the acoustic end of the corresponding spoken unit

	Chinese Example	English Example
1 Filler Removal	Before 那个就是说 有没有 AI Logo 的设计工具， 就是说那个就是说 AI 生成 Logo 的 Polished 有没有 <u>AI 生成 Logo</u> 的设计工具？	Before Um and this means that, um once, uh It it comes together with a voice recognition. Polished <u>And this means that once</u> it comes together with a voice recognition.
2 Repetition Reduction	Before 天天 天天 发 发发 抖音。 Polished <u>天天发</u> 抖音。	Before and uh not all the the the the stuff you can do with it, but the the the essential stuff is there Polished And not all <u>the stuff</u> you can do with it, but <u>the essential stuff</u> is there.
3 Self-Correction	Before 搜一下 遛果香味的红富士苹果， 啊不是，是 烟台栖霞红富士..... Polished 搜一下 <u>烟台栖霞红富士苹果</u>	Before Just to put it just to put it in some comparison; some comparative context. Polished <u>Just to put it in some comparative context.</u>
4 Formatting	Before 要 三 斤装的，还带“脆”字 的 那个。 Polished 要 <u>3</u> 斤装的，还带“脆”字 <u>，</u> 的那个。	Before with chasing ch-r share your daily plan as per your ch-r with your team also Polished <u>With</u> chasing <u>DRR</u>, share your daily plan as per your <u>DRR</u> with your team also <u>。</u>

Figure 13: Representative Chinese and English polishing examples. The model removes fillers and repetitions, resolves self-corrections, and normalizes sentence structure, punctuation, and acronyms while preserving the intended meaning.

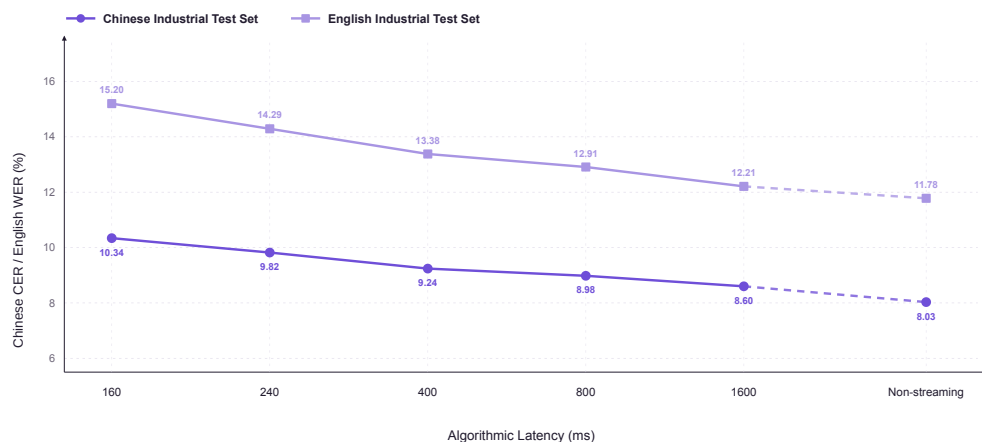


Figure 14: Latency–accuracy trade-off on internal Chinese and English industrial test sets. Chinese CER and English WER decrease as algorithmic latency increases, with the non-streaming setting achieving the lowest error rates.

to the moment that unit first appears in the on-screen partial transcript. First-token latency applies this definition to the first correctly recognized token displayed on screen in each utterance, whereas text-display latency applies it throughout the utterance. As shown in Figure 14, Chinese CER and English WER decrease consistently as algorithmic latency increases, while the non-streaming setting achieves the lowest error rates. These results demonstrate a clear latency–accuracy trade-off and show that the system can be flexibly configured to meet different latency requirements.

Under the low-latency configuration, the system achieves a first-token latency below 200 ms and a text-display latency below 300 ms. At the 400 ms algorithmic-latency operating point in Figure 14,

Table 5: Streaming latency of Doubao Streaming ASR 2.0 and Qwen-Audio-3.1-Asr-Flash-Message. All values are in milliseconds; lower is better.

Metric	Doubao Streaming ASR 2.0	Qwen-Audio-3.1-Asr-Flash-Message
Text-emission latency	530	220
First-character latency	570	100
Last-character latency	307	103

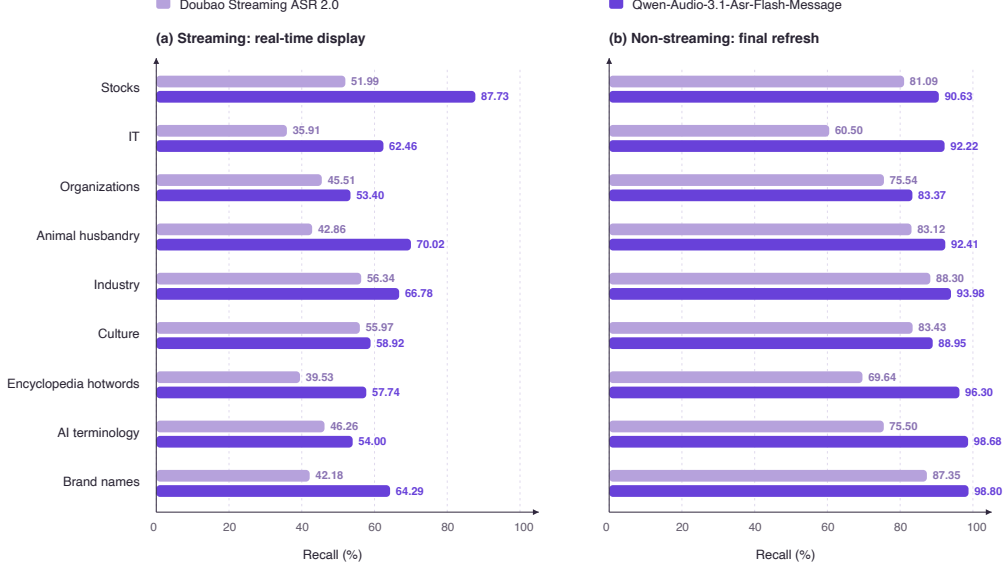


Figure 15: Industry-term recall on nine domain-specific test sets for (a) streaming outputs used for real-time display and (b) non-streaming outputs used for the final transcript refresh. Doubao Streaming ASR 2.0 abbreviates Doubao Streaming Speech Recognition Model 2.0. Each panel compares the two systems within the same output stage. Recall is reported as a percentage; higher is better.

the system achieves a Chinese CER of 9.24% and an English WER of 13.38%, compared with 8.03% CER and 11.78% WER, respectively, in the non-streaming setting. This corresponds to relative error-rate increases of approximately 15.1% for Chinese and 13.6% for English.

5.8 Streaming Display and Non-streaming Refresh in Message ASR

Message ASR combines streaming recognition for real-time transcript display with non-streaming recognition for refreshing the transcript at utterance end. We compare qwen-audio-3.1-asr-flash-message with Doubao Streaming Speech Recognition Model 2.0, abbreviated as Qwen Message and Doubao, respectively. We evaluate streaming latency and industry-term recall separately for the streaming and non-streaming outputs to distinguish the responsiveness and quality of the intermediate displayed text from the quality of the final refreshed transcript.

As shown in Table 5, Qwen Message achieves lower latency on all three reported metrics. Its first-character latency is 100 ms, compared with 570 ms for Doubao, while text-emission latency decreases from 530 ms to 220 ms and last-character latency from 307 ms to 103 ms. These results support responsive real-time transcript display before the non-streaming refresh at utterance end.

For streaming recognition, Figure 15(a) shows that Qwen Message achieves higher industry-term recall on all nine domain-specific test sets. Its unweighted mean recall is 63.93%, compared with 46.28% for Doubao, a difference of 17.64 percentage points. The largest gains occur in stocks (+35.74 points), animal husbandry (+27.16 points), and IT (+26.55 points), indicating improved recognition of domain-specific terms in the intermediate displayed transcripts.

Table 6: Speaker-diarized ASR results on four evaluation sets. Fun-ASR is evaluated as a cascaded system. DER and cpWER are reported as percentages; lower is better. The best result in each column is shown in bold.

System	AISHELL-4		AliMeeting-test		MLC-SLM		MagicData-RAMC	
	DER	cpWER	DER	cpWER	DER	cpWER	DER	cpWER
Fun-ASR (Cascaded)	8.45	21.98	9.03	34.68	6.17	33.83	5.37	20.42
MOSS-Transcribe-Diarize (MOSI)	4.20	16.04	3.38	17.23	3.20	14.06	4.79	14.00
Muse-Voice-Transcribe-1.0 (Meta)	3.04	19.82	6.16	27.97	3.48	14.60	4.55	15.47
Gemini-3.5-Transcribe (Google)	13.99	37.00	21.84	45.16	12.71	23.05	11.44	25.32
Mai-Transcribe-2 (Microsoft)	5.96	23.00	8.52	27.65	7.01	15.63	5.42	14.98
Qwen-Audio-3.1-ASR-Flash	3.23	15.70	2.23	16.00	1.77	12.31	3.04	13.28
Qwen-Audio-3.1-ASR-Flash-Next	2.73	14.17	2.12	11.95	1.74	12.71	3.18	13.63

For the non-streaming refresh, Figure 15(b) shows that Qwen Message again achieves higher industry-term recall on all nine sets, with an unweighted mean of 92.82%, compared with 78.27% for Doubao, a difference of 14.54 percentage points. The largest gains occur in IT (+31.72 points), encyclopedia hotwords (+26.66 points), and AI terminology (+23.18 points). Both systems achieve higher recall after the non-streaming refresh than in their corresponding streaming outputs on every evaluated domain. Together, these results illustrate the complementary roles of the two stages: streaming recognition provides responsive intermediate text, while non-streaming recognition improves industry-term recall in the final refreshed transcript.

5.9 Speaker-diarized ASR

We evaluate joint speaker attribution and transcription on four test sets using DER and cpWER. Table 6 compares the two Qwen-Audio-3.1-ASR variants with a cascaded Fun-ASR baseline and three other systems.

The two Qwen-Audio-3.1-ASR variants collectively achieve the best results across all eight evaluated metrics. Next obtains the lowest DER on AISHELL-4, AliMeeting-test, and MLC-SLM, together with the lowest cpWER on AISHELL-4 and AliMeeting-test. SSE achieves the lowest cpWER on MLC-SLM and the lowest DER and cpWER on MagicData-RAMC. The largest cpWER difference between the two variants occurs on AliMeeting-test, where Next achieves 11.95%, compared with 16.00% for SSE, a reduction of 4.05 percentage points.

Both variants outperform the cascaded Fun-ASR baseline on all four test sets in both DER and cpWER. Next also achieves lower error rates than all non-Qwen systems across all eight metrics. These results demonstrate strong speaker attribution and transcription performance across the evaluated test sets, with the relative strengths of the two Qwen variants depending on the dataset and metric.

5.10 General Audio Understanding

We evaluate Qwen-Audio-3.1-ASR-Flash-Next on seven comprehensive and specialized audio understanding benchmarks measured by accuracy, and two temporal grounding benchmarks measured by mean Intersection over Union (mIoU). Table 7 compares its results with three competing systems.

Qwen-Audio-3.1-ASR-Flash-Next achieves the best performance on seven of the nine benchmarks. On the comprehensive audio understanding benchmarks, it obtains an accuracy of 81.60% on MMAU, which evaluates audio understanding and reasoning across speech, non-speech sounds, and music, outperforming Gemini-3.1-Pro by 0.70 percentage points. On MMSU, which assesses fine-grained spoken-language perception and reasoning, it achieves an accuracy of 82.80%, trailing the best result by 2.50 percentage points.

On the specialized audio understanding benchmarks, Qwen-Audio-3.1-ASR-Flash-Next achieves the best results across all five tasks: acoustic scene recognition (CochlScene), non-speech vocal sound recognition (Nonspeech7k), emotion recognition (MELD), musical instrument recognition (NSynth), and speaker gender classification (DiDiSpeech). The improvements are particularly pronounced

Table 7: Audio understanding results. Accuracy is reported for the first seven benchmarks and mIoU for the two temporal grounding benchmarks; all values are percentages. Higher is better, and the best result in each row is shown in bold.

Benchmark	Gemini-3.1-Pro	Gemini-3.5-Flash	Seed-2.0-Lite	Qwen-Audio-3.1-ASR-Flash-Next
Comprehensive Audio Understanding (Accuracy)				
MMAU	80.90	77.20	78.40	81.60
MMSU	85.30	78.32	81.36	82.80
Specialized Audio Understanding (Accuracy)				
CochlScene	63.90	55.07	49.82	76.30
Nonspeech7k	90.90	76.69	88.83	93.20
MELD	52.40	45.94	57.93	58.12
NSynth	39.20	16.31	29.79	66.90
DiDiSpeech	93.00	64.00	98.33	99.67
Temporal Grounding (mIoU)				
Clotho-Moment	52.56	44.24	71.79	82.20
DESED	49.37	37.86	66.64	66.60

on CochlScene and NSynth, with absolute margins of 12.40 and 27.70 percentage points over the respective next-best results.

For temporal grounding, Qwen-Audio-3.1-ASR-Flash-Next achieves 82.20% mIoU on Clotho-Moment, a benchmark for precise single-interval boundary localization, surpassing the next-best result by 10.41 percentage points. On DESED, which evaluates multi-interval retrieval of recurring acoustic events, it achieves 66.60% mIoU, only 0.04 percentage points below the best result. Together, these results demonstrate robust fine-grained temporal grounding across complementary localization and retrieval settings.

6 Conclusion

In this report, we present Qwen-Audio-3.0-ASR, a large-scale LLM-based automatic speech recognition system built on the Qwen Mixture-of-Experts backbone and trained on tens of millions of hours of speech data. Through a unified instruction-following framework, Qwen-Audio-3.0-ASR supports transcription across 30 languages and 16 Chinese dialectal varieties, together with production-oriented capabilities including industry-domain entity recognition, hierarchical hotword customization, native single-pass transcription polishing, and long-audio contextual modeling. Extensive evaluations demonstrate strong and well-balanced performance across Chinese, English, multilingual, dialectal, and real-world industrial settings, including substantial improvements in long-tail entity recognition, hotword customization, and transcription readability. In addition to Qwen-Audio-3.0-ASR, we develop Qwen-Audio-3.0-ASR-Streaming as a dedicated low-latency streaming model for interactive applications. We further introduce Message ASR as a deployment profile of Qwen-Audio-3.0-ASR for real-world transcription scenarios, integrating historical context, hierarchical hotwords, native polishing, and low-latency transcription within a unified decoding workflow without introducing a separate recognition or rewriting pipeline. Together, these components provide a comprehensive ASR solution for diverse real-world production requirements.

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